SUMALYA BHATTACHARYA

Quantitative Researcher | Systematic Trading, Market Microstructure & Execution Research
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SUMMARY

Quantitative researcher with **3 years at a hedge fund** developing systematic equity strategies and statistical-arbitrage models, running the full research cycle from data engineering and hypothesis design through walk-forward validation, risk sizing and transaction-cost analysis. Strategies reached **out-of-sample Sharpe 1.3–1.8 with max drawdown < 12%** in walk-forward backtests. Independent research on **~1.3B crypto tick trades** produced a maker-fill selection signal worth **+5.1 bps incremental mark-out** on an untouched 14-day holdout, costed against queue position and exit fees. Disciplined about look-ahead bias, multiple testing and keep/drop decisions, including dropping hypotheses that reverse out of sample. Python, C++, SQL, q/kdb+.

PROFESSIONAL EXPERIENCE

DMNS Network Pte. Ltd. — hedge fund **Oct 2023 – Present**
Quantitative Trading Analyst (joined as Quantitative Research Intern)
Ran the full research cycle on Indian equity and multi-asset strategies: data pipelines → hypothesis → walk-forward backtest → risk sizing → reproducible, versioned signal code.

Systematic Trend & Regime Strategies — NIFTY50, medium frequency
Python, Pandas, NumPy, scikit-learn, Backtrader

- Developed medium-frequency trend-following and regime-switching strategies reaching **out-of-sample Sharpe 1.3–1.8 with max drawdown < 12%** in walk-forward backtests; parameters were chosen only inside training windows so every test period stayed untouched.
- Raised trade precision **15–20% in high-volatility regimes** by gating breakout entries on the implied-minus-realised volatility spread, filtering out false breakouts.
- Built the data and code base behind every result: OHLCV ingestion that repairs missing sessions, stock splits and duplicate timestamps, and Git-versioned signal modules so any backtest can be reproduced exactly.

Statistical Arbitrage & Risk Management — crypto and ETF pairs
Python, Statsmodels, SciPy, Kalman filter, copulas, Parquet

- Built a mean-reversion engine that commits capital only to fast-reverting spreads, screening pairs by cointegration (ADF) and **Ornstein–Uhlenbeck half-life**.
- Estimated **time-varying hedge ratios with a Kalman filter** and modelled non-linear tail dependence between legs with Student-t, Clayton and Gumbel **copulas**.
- Designed the risk layer: volatility-targeted, **half-Kelly** position sizing with circuit breakers on drawdown, leverage and regime breaks; multi-timeframe ingestion fills weekend gaps without look-ahead bias.

INDEPENDENT RESEARCH

Liquidation-Cascade Signal & Maker-Fill Filter — BTC/ETH perpetual futures **2026**
Binance + Bybit tick data, Dec 2025 – Feb 2026 | Python, PyArrow, Parquet, NumPy, SciPy, LightGBM

- Built ~80 strictly causal order-flow, price-displacement (**IC ≈ 0.14**) and cascade-intensity features (Hawkes intensities, forced-flow VPIN, cross-asset lead–lag) on **~1.3B tick trades** through a streaming PyArrow/Parquet pipeline; found Binance's liquidation feed is throttled to 1 row/s per symbol and re-specified it as a sign/timing signal, taking size from Bybit.
- Selected maker fills with **+5.1 bps higher 120 s mark-out than the unfiltered fill set** (kept fills vs. all fills) on a frozen-spec, one-shot **14-day holdout**; day-bootstrap 5% lower bound +2.4 bps, 82% of coin-days positive, random placebo ≈ 0.
- Ran a model ladder (threshold rules → ridge → monotone GBM → LightGBM) on identical 28-day walk-forward folds with 1-day embargo, Bonferroni and Hansen SPA controls; chose a **3-feature composite that beat LightGBM out of sample (5.1 vs. 3.4 bps)**, pruned 39 features via a pre-registered keep/drop ledger and dropped a side-asymmetry hypothesis that reversed sign out of sample.
- Tested tradability with a **queue-position and exit simulator** on a 1 s book panel: break-even taker exit cost 5.2 bps at 120 s; the edge holds after a 2 bps taker exit (+3.2 bps net, positive lower bound) and with passive exits (+5.6 bps, 78–86% fill at a 10 s timeout), but is gone by 300 s, so viability depends on fee tier and a short holding horizon.

Options Analytics Microservice — NSE index options
Python, FastAPI, WebSockets, React, Docker, Nubra SDK | github.com/sumalya41/nubra-options-microservice-SB

- Streamed live **Greeks and implied volatility with sub-second latency** through an asynchronous FastAPI/WebSocket microservice, with TradingView charts in React dashboards.
- Made the **IV surface** reliable by rejecting missing strikes, stale quotes, bid-ask anomalies and timestamp mismatches before calibration; containerised with Docker in a CI-ready repository.

Chinese Commodity Futures — SHFE / DCE / CZCE
Python, Pandas, NumPy, Statsmodels

- Built a **continuous-contract dataset across 77 futures contracts** for intraday and overnight exposure research, with open-interest / volume roll rules and back-adjustment so returns carry no artificial roll gaps.

ADDITIONAL RESEARCH PROJECTS

Hierarchical Risk Parity & Backtesting Framework — 50+ US equities
Python, NumPy, SciPy, Pandas | github.com/sumalya41/Hierarchial-Risk-Parity

- Implemented **HRP with generalised-hyperbolic covariance** in a modular object-oriented framework and tested it on a **20-year out-of-sample walk-forward** (Sharpe, drawdown, risk decomposition).
- Compared HRP with inverse-variance and mean-variance weights on identical rebalance dates, using risk-contribution decomposition and drawdown attribution to show how risk is shared across clusters.

Black-Litterman Portfolio Optimisation
Python, NumPy, SciPy, Pandas | github.com/sumalya41/QFin.Colab

- Cut portfolio volatility **10–15% vs. equal weight** at higher risk-adjusted return by combining Markowitz mean-variance optimisation with **Black-Litterman** views.
- Added allocation constraints, volatility targeting and dynamic rebalancing with SciPy optimisers; stabilised weights with outlier-filtered, rolling-normalised covariance estimates.

Advanced Portfolio Construction — CPPI & risk budgeting (EDHEC framework)
Python, Statsmodels, SciPy | github.com/sumalya41/EDHEC_BS_2.Py-ML-Advanced-Portfolio-Construction-and-Analysis

- Built **CPPI** capital protection and dynamic asset allocation to cap downside exposure under non-linear market conditions.
- Engineered multi-factor **risk budgeting** with volatility- and correlation-driven allocation, keeping cross-asset exposure stable as regimes changed.

Energy Market Analytics — oil & gas
Python, Statsmodels, Pandas, FRED API | github.com/sumalya41/OIL_GAS_Analysis

- Built statistical forecasting pipelines on oil and gas prices and volumes to generate macro-risk and commodity-trading signals; outlier filtering and feature normalisation made forecasts more stable.
- Modelled supply–demand dynamics with time-series methods, combining FRED macro data with alternative data to study event-driven volatility and hedging against sudden market shocks.

Distributed Market-Event System — Erlang/OTP
Erlang/OTP, Mnesia, BEAM VM | github.com/sumalya41/erlang_Contracts

- Built fault-tolerant, actor-based concurrent services under OTP supervision trees for high-throughput market-event processing, with Mnesia-replicated transactional state and deterministic state machines for deadlock-free parallel execution; modelled financial contracts in the composable-contracts style (Peyton Jones et al.).

TECHNICAL EXPERTISE

Programming: Python (advanced), C++ (Boost), SQL, q/kdb+, MATLAB, Erlang/OTP

Quantitative research: alpha research, market microstructure, statistical arbitrage, time-series modelling, Hawkes processes, Kalman filtering, copulas, regime detection, portfolio optimisation, option Greeks and IV surfaces

Research validation: purged walk-forward CV with embargo, day-block bootstrap, multiple-testing control (Bonferroni, Hansen SPA), look-ahead bias tests, transaction-cost and queue/fill simulation

Data & infrastructure: tick-data stores (zstd Parquet, PyArrow streaming), NumPy, Pandas, SciPy, Statsmodels, scikit-learn, LightGBM, Kafka, PySpark, FastAPI, Docker, Git, Linux

Markets & data: crypto perpetuals (Binance, Bybit), NSE equities and index options, Chinese commodity futures, US equities; Bloomberg, FactSet

EDUCATION & ACHIEVEMENTS

M.Sc. Data Science — Manipal University **2024 – 2026**
Thesis: market microstructure — spread decomposition, Kyle's λ price impact, Almgren–Chriss optimal execution.

M.Sc. Applied Mathematics (Minor: Financial Mathematics) — Parul University **2023 – 2025**

Post Graduate Diploma in Applied Statistics — Indian Statistical Institute, Kolkata **2023**

B.Com (Hons) Finance — University of Calcutta | **CA Intermediate (IPCC)** — ICAI

Achievements: top 10th percentile, IIT Kharagpur Finance Club Trading Competition 2025; poker market-making games (EV-based quoting and sizing).

Courses: Market Microstructure — Università di Napoli Federico II (2026); Behavioral Finance — Duke University (2025).

Languages: Bengali (native), English (fluent), Hindi (conversational).